4.2.20

EE25BTECH11017 - Mahesh Chollangi

Question:
Find the direction and normal vector for the line

$$x + 2y = 6 \tag{0.1}$$

Solution:

This equation can be expressed in terms of matrices
Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{0.2}$$

$$\mathbf{n}^{\mathbf{T}} = \begin{pmatrix} 1 & 2 \end{pmatrix} \tag{0.3}$$

$$c = 6 \tag{0.4}$$

The line equation can be written as:

$$\mathbf{n}^{\mathbf{T}}\mathbf{x} = c \tag{0.5}$$

Where $\mathbf{n}$ is the normal vector of the given line
The direction vector of the line can be found by observing the normal vector.

$$\mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \tag{0.6}$$

This is true because if the director vector is represented as

$$\mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix} \tag{0.7}$$

then the normal vector can be represented as

$$\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix} \tag{0.8}$$

This can be verified by the following equation:

$$\mathbf{n}^{\mathbf{T}}\mathbf{m} = 0 \tag{0.9}$$

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 0 \tag{0.10}$$

The normal vector of the line is $\mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ The director vector of the line is $\mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

From the figure, it is clearly verified that the theoretical solution matches with the computational solution.